

Research Notes on ResNet deep learning

ResNet deep learning

$$y = F(x, \{W_i\}) + x$$

>> Section 1

Neural networks have been used for implementing language models since the early 2000s. LSTM helped to improve machine translation and language modeling. Other key techniques in this field are negative sampling and word embedding. Word embedding, such as word2vec, can be thought of as a representational layer in a deep learning architecture that transforms an atomic word into a positional representation of the word relative to other words in the dataset; the position is represented as a point in a vector space. Using word embedding as an RNN input layer allows the network to parse sentences and phrases using an effective compositional vector grammar. A compositional vector grammar can be thought of as probabilistic context free grammar (PCFG) implemented by an RNN. Recursive auto-encoders built atop word embeddings can assess sentence similarity and detect paraphrasing. Deep neural architectures provide the best results for constituency parsing, sentiment analysis, information retrieval, spoken language understanding, machine translation, contextual entity linking, writing style recognition, named-entity recognition (token classification), text classification, and others. Recent developments generalize word embedding to sentence embedding. Google Translate (GT) uses a large end-to-end long short-term memory (LSTM) network. Google Neural Machine Translation (GNMT) uses an example-based machine translation method in which the system "learns from millions of examples". It translates "whole sentences at a time, rather than pieces". Google Translate supports over one hundred languages. The network encodes the "semantics of the sentence rather than simply memorizing phrase-to-phrase translations". GT uses English as an intermediate between most language pairs.

>> Section 2

Weather prediction Traditional weather prediction systems solve a very complex system of partial differential equations. GraphCast is a deep learning based model, trained on a long history of weather data to predict how weather patterns change over time. It is able to predict weather conditions for up to 10 days globally, at a very detailed level, and in under a minute, with precision similar to state of the art systems.

>> Section 3

A large percentage of candidate drugs fail to win regulatory approval. These failures are caused by insufficient efficacy (on-target effect), undesired interactions (off-target effects), or unanticipated toxic effects. Research has explored use of deep learning to predict the biomolecular targets, off-targets, and toxic effects of environmental chemicals in nutrients, household products and drugs. AtomNet is a deep learning system for structure-based rational drug design. AtomNet was used to predict novel candidate biomolecules for disease targets such as the Ebola virus and multiple sclerosis. In 2017 graph neural networks were used for the first time to predict various properties of molecules in a large toxicology data set. In 2019, generative neural networks were used to produce molecules that were validated experimentally all the way into mice.

>> Section 4

Commercial activity Facebook's AI lab performs tasks such as automatically tagging uploaded pictures with the names of the people in them. Google's DeepMind Technologies developed a system capable of learning how to play Atari video games using only pixels as data input. In 2015 they demonstrated their AlphaGo system, which learned the game of Go well enough to beat a professional Go player. Google Translate uses a neural network to translate between more than 100 languages. In 2017, Covariant.ai was launched, which focuses on integrating deep learning into factories. As of 2008, researchers at The University of Texas at Austin (UT) developed a machine learning framework called Training an Agent Manually via Evaluative Reinforcement, or TAMER, which proposed new methods for robots or computer programs to learn how to perform tasks by interacting with a human instructor. First developed as TAMER, a new algorithm called Deep TAMER was later introduced in 2018 during a collaboration between U.S. Army Research Laboratory (ARL) and UT researchers. Deep TAMER used deep learning to provide a robot with the ability to learn new tasks through observation. Using Deep TAMER, a robot learned a task with a human trainer, watching video streams or observing a human perform a task in-person. The robot later practiced the task with the help of some coaching from the trainer, who provided feedback such as "good job" and "bad job".

>> Section 5

Relation to human cognitive and brain development Deep learning is closely related to a class of theories of brain development (specifically, neocortical development) proposed by cognitive neuroscientists in the early 1990s. These developmental theories were instantiated in computational models, making them predecessors of deep

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learning systems. These developmental models share the property that various proposed learning dynamics in the brain (e.g., a wave of nerve growth factor) support the self-organization somewhat analogous to the neural networks utilized in deep learning models. Like the neocortex, neural networks employ a hierarchy of layered filters in which each layer considers information from a prior layer (or the operating environment), and then passes its output (and possibly the original input), to other layers. This process yields a self-organizing stack of transducers, well-tuned to their operating environment. A 1995 description stated, "...the infant's brain seems to organize itself under the influence of waves of so-called trophic-factors ... different regions of the brain become connected sequentially, with one layer of tissue maturing before another and so on until the whole brain is mature". A variety of approaches have been used to investigate the plausibility of deep learning models from a neurobiological perspective. On the one hand, several variants of the backpropagation algorithm have been proposed in order to increase its processing realism. Other researchers have argued that unsupervised forms of deep learning, such as those based on hierarchical generative models and deep belief networks, may be closer to biological reality. In this respect, generative neural network models have been related to neurobiological evidence about sampling-based processing in the cerebral cortex. Although a systematic comparison between the human brain organization and the neuronal encoding in deep networks has not yet been established, several analogies have been reported. For example, the computations performed by deep learning units could be similar to those of actual neurons and neural populations. Similarly, the representations developed by deep learning models are similar to those measured in the primate visual system both at the single-unit and at the population levels.